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# Digital Publisher, Advertising Media Agency and Mobile Exchange Triadic Interaction: Digital Marketing Service Supply Chain Landscape in Indonesia

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***Abstract*—The purpose of this paper is to describe the digital marketing supply chain landscape in Indonesia, an emerging market country with high economics and e-commerce penetration growth. The scope of this paper will discuss structures, actors, business model and driving forces of digital advertising with the supply chain context, which will also explain, information, money and product/services flow. This research is using mix-method approaches, combining literature review, industry data analysis through desk study and experts' in-depth interview. There are three main findings, 1) mobile exchange company through its mobile exchange platform creates price transparency for paid digital advertisement, 2) there is power imbalance between digital publishers, advertising media agencies and mobile exchange, 3) If there are nothing significant strategic management changes in advertising media agency side, mobile exchange company could take out the (digital) advertising media agency within marketing communication industry. This paper is expected to give more insights for practitioners which include advertisers, advertising agencies, digital publishers as well as academician, especially on the digital marketing supply chain landscape. The originality of this paper lies in the digital advertising media landscape, mobile exchange business model with supply chain context, which have not been mapped to date. This study still on the exploratory approach and only focus in paid advertising, not including social media. Therefore, in order to get more whole-rounded point of view in future; first, more experts informants from different type of actors are needed, especially advertisers from FMCG company, management of digital publishers, management of social media platform, digital key opinion leaders, etc.; second, solid empirical future research will create better understanding toward the industry landscape and supply chain in different context.***

***Keywords*—component, formatting, style, styling, insert**

## I. Introduction

Indonesian Advertising Agency Association (PPPI) reporting that digital media advertising spending has grown up to approximately 15% per annum and at the same time digital media industry has gone through a fantastically rapid transformation by the presence of digital technology that five impact and consequences to the companies and their consumers [1]. The establishment of "mobile-first"-based economic evolution and hyper-social behavior of Indonesian consumers are the key influencing factors of digital-mobile media advertising development in Indonesia.

PPPI also reporting that Internet access through smartphones has changed the procedure and society's point of view either on people's personal or professional lives. Whenever and wherever they want, at an appropriate time, Indonesian consumers will use smartphones and other devices as primary computer access to information. Smartphone penetration has grown up more than 24% of total cellular phone customers in Indonesia due to the lower cost of having a smartphone. Internet data usage of big 3 telco players in Indonesia (Telkomsel, XL Axiata and Indosat) has grown twice as much as that in 2011 and it was even predicted to be grown up 5 times at the end of 2018. Indonesian population also enjoys the presence of apps (mobile applications) in their cell-phones and slowly leaves "web browsers". Optimization of "mobile-web" becomes very important and requires bounteous attention from advertisers where smart-gadget domination pushes advertising agencies to provide the best inter-device experience, inter "operating system" and even manage appearance difference in the screen size [2].

PPPI's finding is in-line with publication from WeAreSocial 2017 report [1]. Indonesia, with total population more than 262 million people (est.2016), 55% urbanization rate, have 132,7 million internet users (51% penetration, increasing 45 million since January 2016), 106 million active social media users (40% penetration, increasing 27 million since January 2016), 371,4 million mobile subscription (142% penetration, increasing 45 million since January 2016), and 92 million active users (35% penetration, increasing 26 million since January 2016). This could be concluded that Indonesia is indeed one of the highest potencies to become digital and mobile media players worldwide.

There are also interesting facts from the Indonesia Association of Internet Service Provider (APJII) 2016 report. First, most Indonesian connect the internet through the mobile platforms (92,8 million users) compare to 17,7 Million accessing from home, 14,9 million accessing from campus, 2,9 million from internet Café and 1,2 million from Café. This fact is implying a strong message that the mobile platform is the most significant media platform, as well as advertising media device that use digital technology, compare to laptop or computer devices. Second, most reasons to connect the internet search for information 31,3 million, work-related 27,7 million, leisure 17,9 million, socialize 13,6 million, entertainment 11,7 million, education 12,2 million and trade 10,4 million [3]. This fact implies that the shift from the classic marketing communication AIDA (awareness-interest-desire-action) approach to AISAS (Awareness-interest-search-action/download-share) approach is confirmed for the Indonesian consumer.

Meanwhile If we look at other facts from Globalwebindex Q3 & Q4 2016 through WeAreSocial 2017, in regards of average daily time spent with Media for internet users aged 16-64, Indonesian tend to use 8 hours and 44 minutes on their average daily use of internet via PC or tablet, 3 hour 55 minutes daily use of internet via mobile phone, 3 hours 16 minute of social media and 2 hours 23 minutes television viewing time.

Based on Techinsasia 2016 research-based prediction, Indonesia e-economy sector is predicted to grow dramatically in 2025. E-commerce as the prime driver of e-economy sectors will be 22 times higher than today. Based on a survey by emarketer.com, 60% of internet users are digital natives and 71% of internet users in Indonesia said that they purchased a product after watching influencer's advertisement. Accordingly, it could be inferred that e-commerce plays a significant role not only as a distribution channel but also a strong presence of digital media.

Similar to Techinasia research, business-insider also finds that Indonesia by 2020, is projected to reach US$130 Million, and making Indonesia the 3rd largest e-commerce market behind China and India. In 2025, Southeast Asia's digital economy (which includes e-commerce, online games, and online advertising) will surge to USD 200 billion. And Indonesia forecasted to have accounted for 40.5% or USD 81 billion of its total market region, and half of it comes from Indonesian e-commerce. In 2016, third party merchants (Lazada, Tokopedia, Rakuten, Elevenia, etc.) led internet retailing in Indonesia by 34% of value sales. Blibli.com recorded as the highest increase in terms of value sales in internet retailing in 2016.

There are some interesting facts about consumer behavior of e-commerce in Indonesia based on *Globalwebindex* Q3 & Q4 2016 data, as reported in WeAreSocial 2017. The survey informs that 48% of the respondents use web for searching product/service to buy, 46% visited an online retail store, 41% purchased a product or service online, 34% made an online purchase via a laptop or desktop computer and 33% made an online purchase via a mobile device. *Ecommercein.asia* also reports that 39% of Indonesian e-commerce consumers start their product journey online through forum (Kaskus), Social Media channels (Facebook/Instagram), search engine (google), product review, brand or retailer and marketplace. These results are also implying that there is a strong connection between digital media usage and e-commerce performance.

While the growth is promising, WeAreSocial 2017 report also describe that total value of the consumer (B2C) e-commerce market and average e-commerce revenue per user per January 2017 in Indonesia still has room for growth. Number of people purchasing via e-commerce 24,74 million, e-commerce penetration (number of purchaser vs total population) 9%, Total value of national e-commerce market in 2016 US $ 5,6 billion (compare to nearly US $ 1 trillion economy) and average annual e-commerce revenue per user in 2016 US $ 228 (compare to US $ 3,603 per capita).

Based on our preliminary expert in-depth Interview with digital marketing practitioners, there are three main actors with one minor actor within the digital marketing landscape, as the main communication channel to connect the consumer with an e-commerce business. They are Advertising media agency, digital publisher, mobile exchange and ad-network

Advertising media agency is a business entity that becomes a middleman between advertiser/brand owner with media/publisher in conducting advertisement inventory sales from media. The business model of advertising media agency is relying on service commission from clients and discount rates from media. Most (traditional) media will give discount to advertising media agency since advertising media agencies aggregate the demand through their list of clients for the media buying process, therefore it increases bargaining power toward media/publisher.

Digital publisher is a publishing site/platform that produces articles, information that specifically targeted a certain type of readers through internet-based connection. Digital publishers have flourished due to internet accessibility and consumer shifting behavior toward media consumption from print-media to the internet. The presence of digital publishers could be view in laptops, tablet and mobile phones. Digital publisher advertisement inventories have an almost infinite 24/7 business model compare to traditional media (TV, radio and print) due to the unlimited number of page-sites, sites-size, and technology used. meanwhile TV and radio advertisement inventory which relatively finite with the limited number of placement of magazine pages or specific commercial air-time. However, Since the digital media have cluttered issue (the amount of media is high, but the viewers tend to be specifically segmented), its bargaining power, especially toward advertising media, is relatively low.

Mobile exchange is a platform where buyer and seller of digital publisher inventory have a more transparent pricing transaction through mobile exchange system that uses Real-time bidding process, a process which operated similarly with the stock-exchange market. Mobile exchange through its server/IT infrastructures, transparent real-time bidding, advance "consumer programmatic buying" programming, networks of digital publishers and big data consumer analytics capacity, is offering more value-added services compare to advertising media agency "business as usual" in capturing digital publishers and advertisers. By using demand-side system platform, supply-side system platform and agency trading desk, Mobile exchange becoming an exclusive connector between digital publishers, advertisers and advertising media agencies. It could be inferred that Mobile exchange service has created a specific e-commerce service that have unique business model which have the potencies to disrupt current business of advertising media agency since it has direct system connection with digital publishers and advertisers.

The minor player, called ad-network, is an aggregator of digital publishers with lower hit numbers and efficiency. This ad-network serve the specific needs on digital publisher and advertising media agency especially on efficiency on the bottom of the pyramid customers that still do trial and error and/or lower budget in digital marketing. Ad-network also has an agreement with digital publishers but much lower in number compared to mobile networks that possess more than 95,000 digital publisher networks. Besides the quantity, the quality of ad-networks digital publishers is also lower compared to the mobile-exchange.

Understanding toward Media advertisement platform is required to understand the range of digital marketing services as well as interaction among digital publishers, advertisers, advertising media agency and mobile exchange. We could see the common types of digital/mobile media advertising platforms in Indonesia in Table I [4].

TABLE I. COMMON TYPES OF DIGITAL/ MOBILE MEDIA ADVERTISEMENT PLATFORMS IN INDONESIA

| Type | Description |
|---|---|
| Native advertise-ment | Type of advertisement that its display customized to engage with customer experience. This advertisement is created by companies/brands, media or publishers. The purposes of this type of advertisement are to develop/create engagement advertisement communication message to the customer as well as increasing advertisement relevance and effectiveness |
| MRAID | A specification written by interactive advertising bureau for mobile advertising. A set of specific commands that specifically customized to work with java-script and HTML 5 language, which developers could create Rich Media advertisement that can use to communicate with the apps they are being served into. |
| Pre-roll | A mobile advertisement clip streaming before/prior to mobile video/TV clip. This advertisement type last 10-15 seconds with 5 seconds uninterrupted pre-roll show. “Youtube” is a platform that mostly use Pre-Roll advertisement |
| Rich Banner | Advertisement format that are engaging and interactive with broad scope, that include embedded video, embedded audio an expandable banner |
| Off-Deck | Point of access within mobile network, where customer can purchase/access mobile content, products, utilities and information. Location of the access is outside the carrier’s portal/deck |
| Loc-based advertising (LBA) | Advertisement within mobile platform that using specific geographic location coordinates (latitude/longitude) of a specific mobile user |
| Re-target | An approach that utilize consumer browsing habit information to re-send advertisement(s) to the consumers on a different publisher’s website. This approach is done through pixel tagging or other coding that make a publisher to recognize specific users |
| Pixel Tag | A technique that marketers used to track which consumers are clicking through advertisement and trafficking to landing pages by placing an object on page that hold information and delivers the information about the device to website host |
| Site tagging | Tactic on inserting advertisement tags into mobile website (WAP) that enable mobile campaign platform to deliver advertisement to the site |
| Geo-target | Displaying advertisement to consumers based on consumer mobile device location. Could be done by using registered ZIP code information or GPS coordinates that collated by an application/site |
| Freq. cap | The maximum number of an impression that could be exposed to advertisement message |
| Conversion track | An instrument that make advertisers could see/observe how the interaction between consumers and their brand within the marketing funnel. Advertisers could describe traceable events on (mobile) websites or within mobile applications to examine consumer engagement and/or direct response campaigns impact |
| Lat/ Long | The location data (latitude/longitude) that can be shared from (mobile) device, which derived from geographic coordinate system that used in the context of mobile advertisement, to assist the identification of user characteristics, environment or context |

This paper is focusing on understanding the triadic interaction within digital marketing supply chain member network especially in emerging markets with the case study of the Indonesia digital marketing landscape. This paper will be presented sequentially from the literature review, conceptual models, findings, conclusion, limitation and future research.

## II. LITERATURE REVIEW

Technology advancement had created disruption that changed the form of media and the way consumers consume the media [5], [6]. Anyone can see the emergence of digital television, online news, and social media among others that start to replace tradition media such as radio and print media. Moreover, mobile technology advancement through mobile phones had developed specific consumer/viewer behavior on consuming media. Media are becoming not only as close as fingertips to the readers but also easy and much affordable to produce.

In-line with researches that focus on how consumer consume the media, [7] conduct survey research to learn on younger individual consumers motivation toward internet, Facebook and internet advertising. The survey is divided into two studies, first conducted to learn young individuals’ motivations toward Internet, Facebook, and their attitudes and behaviors toward Internet advertising, and second how behaviors toward Facebook advertising in addition to Facebook feature use.

From [7] first study, results showed that structure time, peer influence and quality of life predicts both the use of one-to-many communication features (like FB groups and status update) as well as one-to-one communication feature (like chat and private messenger). FB groups are more incline into customers who are motivated by structure time, peer influence and quality of life. Meanwhile, comments affecting security in privacy concern. The more users feel secure and less privacy concern, the more they wrote comments on FB.

Similar to [7], [8] find that Facebook, through leveraging viral marketing and word of mouth, it could increase their brand equity and brand image, and at the end enhancing consumers’ intention to purchase. [8] also underline that the advertising phenomenon has shifted to trust-based advertising instead of push advertising, and social media like Facebook is a good proxy of trust-based advertising platform. However, despite FB provide a trust-based approach, it should include features like personalization, participation, feedback and realization. Many FB users acknowledge that they have higher willingness to buy after seeing the most “like’s” and “share’s” which implies that the brand is reputable. FB users, which dominated by young adults tend to search for information on the brand’s pages by the recommendation and practices of their close networks and friends. As a conclusion, [8] emphasize that to survive in current fierce market competition, corporations need to be informed of their consumers’ brand awareness as well as devise the current practice of advertising strategies.

The previous studies also confirmed by [9], that mention Digital media growth correlates with the growth of e-commerce. Digital media tends to develop faster when e-commerce activity is high. E-commerce volume is an indicator of people’s propensity to adopt digital technology. It may inform us how digital technology and media has become an important part of people’s lives.

Meanwhile [10] make emphasis that there is a significant difference between the traditional decision processes and mobile decision processes among consumers (Fig. 1). Traditional decision process more lateral process, while mobile decision process has a circular process with each process step could influence each other. [10] also reminds that retailers are becoming aware, since “consumers don’t look at online and in-store as different channels"; instead, consumers tend to focus on solutions to their individual needs and aspirations, which often happen in various step or channel within their shopping process. There are 4 building blocks of Consumer decision process according to [10]: 1) Consumer-retailer inter-connectedness, 2) Consumer empowerment, 3) Proximity-based consumer engagement, 4) Web-based consumer engagement.

To increase the omnichannel marketing power, retailers could utilize collected information from consumers for equipping and empowering the store networks to influence the consumer's in-store decision process and their out-of-store shopping behavior.

In regards to omnichannel shopping, [11] explain that the significantly increasing of smartphone as broadband mobile communication, provide advertisers an opportunity to offer personalized/customized advertisement to its consumers. The personalized/customized approach not only could improve resource efficiency to meet customer needs and aspirations, but also improve the negative sentiment of smartphone subscribers. As consequence, advertising effectiveness will increase especially in creating new business opportunities for the products and service they offer. However, to maximize marketing effectiveness, different key attributes may be required in creating a personalized advertising message.

[11] also mention that to help increase mobile advertisers' effectiveness in delivering mobile advertisement, it required a fusion between personalized advertisements and interactive broadband mobility service. As consequence, it will create a new business paradigm for the mobile device industry. [11] also suggests that collaboration to collect enough information and design is a must, for targeting the right consumers at the right time in the rate place. The ecosystem and roles of advertisements are discussed to enhance the consensus in developing a model of collaboration.

Aligned with [11], [12] explain that the usage of phones empowers consumers in enhancing their competence and connecting consumers to retail scape essentially on the consumers' preferences. However, the increasing interconnectivity provided by smartphones could be problematic for consumers. One of the findings is consumers were reported being anxious by mobile-phone driven shopping possibilities. The continuous connection to news stream creating what so-called fear being "out of the loop" condition. A big part of consumers also concerned with the fact that they could shop more than they needed because it is easiness to do transactions for impulse desire. However, consumers realized that stress and anxiety resulted from technical qualities, must be managed through a various approaches that focuses on regaining self-control and reducing mobile phones' influence on consumers' lives.

[13] try to highlight while mobile shopper marketing, from planning and execution, is tremendously growing, many aspects still need to be examined. His research has raised the contextual issue of mobile shopper marketing scope and the focal concerns to marketers with regards to mobile shopping identification of main entities (shoppers, organization, mobile technology, and employee).

[14] attempt to highlight key qualities related to online/mobile/e-commerce promotions. Firstly, frequent and advance users of smartphone and online shoppers tend to have more e-commerce shopping applications on their phones. Second, understanding on the number of shopping applications owned by consumers could be examined through browsing breadth and frequency of non-shopping applications. Third, transactions created from shopping applications are understood by user browsing information and digital experience. Fourth, consumer digital experience is associated with online retail apps possession than traditional retail. Fifth, transactions through online retailer applications are could be explained by browsing behavior and digital experience.

Despite what [8] emphasizing that the phenomenon of advertising has changed from push advertising to trust-based advertising, [14] explain the existence of push and pull strategy still relevant

Lastly, the study of [15] underlining mobile/digital advertising monetizing issues that driven by concerning facts that most of the existing studies have more focus on social media, undermining paid advertising content. This condition caused by social media is considered interactive, fun and personalized which any direct corporations' interventions and/or commercial activities will be considered negative experiences by the user. Despite the condition might be logically correct, the analysis of revenue stream for social media showing that paid (corporation) content are top of the revenue list. In other words, companies are willing to pay content on social media because it could achieve the desired business expectation, probably due to could targeting users at the right time and the right place like the characteristics of social networks.

## III. Research Method

We propose our conceptual model based on Academic journal articles, industry reports, and experts interviews. Indepth Interview with experts which consists of two persons with Point of discussions consists of 1) Advertising Media agency industry landscape, 2) Digital marketing landscape in Indonesia, 3) Mobile exchange business model & services and 4) Ad-exchange business model and what aspects that make it different with Mobile exchange. The two experts are:

1. Mrs. C.A – Until January 2019, she's C level executive of Mobile exchange Indonesia, with more than 18 years of experience in marketing service industry which include advertising agencies and fast-moving consumer goods companies and mobile exchange.
2. Mr. RP – Until January 2019, he's Grouphead Planning in the biggest advertising media agency conglomeration in Indonesia both billing and number of clients. He has more than 15 years of experience ranging in advertising media agencies, marketing research agencies and Television network

The business model proposition of Mobile exchange is creating price transparency between digital publisher and the buyer especially end-customer the advertisers. This approach is taken due to the empirical findings that grey pricing conducted by advertising agency could lead what so called economics inelastic pricing loss due to no clear pricing both between digital publisher and advertising media agency as well as advertiser with advertising media agency. In order to have clear pricing within the supply chain network, mobile exchange facilitate it through mobile exchange with bid ask pricing system that similar with stock exchange real time bidding system. Mobile exchange also has the capability of data processing which size and security level almost equal to capital market – In Indonesia mobile exchange operate its business with similar regulation required by stock-exchange. Mobile exchange in Indonesia posses strong alliances with digital publishers (more 90,000 digital publishers worldwide). The matching transaction within mobile exchange, start from the lowest price offered by the digital publisher with the highest price bid by buyers (could be advertisers or advertising media agency) meanwhile mobile exchange will received the transaction fee.

The waterfall system explaining that the business of mobile exchange is not only doing trading exchange with publisher inventory through real time bidding systems. It also deals with pricing strategy regime in order to optimize the profitability from the business model. The logic behind is each page sites have different value due to number page hits and unique view. A first landing pages of website have more high value compare sub section page site which need double access. However, due to different type of customers that have specific segmentation as well as consumer big data analytics capacity, mobile exchange could offer specific service that match the client target customers with specific sites that match its clients aspirations.

Advancement of technology could make mobile exchange could create solution if the competitors or other companies has booked out the preferred pages. For example Oppo have booked out all the first landing page sites of Indonesian big portals like detik.com, kapanlagi.com, for months. Meanwhile, Samsung have the similar customer aspiration and target consumers with Oppo, which the one month- fully booked out condition make Samsung have less share of voice presence compare to Oppo. Through digital consumer analytics, Mobile exchange could provide information and redirect consumer who has access Oppo advertisement and make the next consumer landing page have Samsung advertisement.

Mobile exchange Indonesia have price waterfall scheme method as part of product development and pricing strategy. The top level of the waterfall consists of the the preferred deal and private market place which has higher price per CPM (cost per mil) or cost per 1000 click. The top level representing 20% of total publisher inventory which most likely higher value page sites. The next level is Media banner, the classic digital advertisement that could lead to specific sites. The bottom of the level are ad-networks and other content providers which present a lower end sites which little page value and/or lowest page hits and/or sites with content that not preferred my most of brands like online gambling sites, pornographic/ sexual-related content sites, etc..

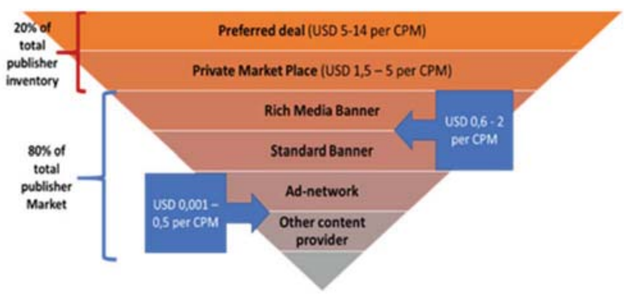


Fig. 1. Mobile exchange "price waterfall" scheme

We also found that advertiser understanding about mobile exchange and digital marketing landscape will lead to effective and efficient digital marketing strategy for digital paid marketing. One of Mobile exchange marketing strategy is having constant training & educational program toward digital marketing and mobile exchange. The training is targeting digital marketing function within business organization which also include the advertising media agency on how to conduct digital marketing transaction with mobile exchange by using agency trading desk and demand side platform.

Table II describe the differences of mobile exchange and other digital publisher aggregator especially functions that could not be fulfilled by non-mobile exchange service (i.e. : ad network) in conducting digital publisher inventory transaction as part of advertisers digital marketing campaign [4].

TABLE II. MOBILE EXCHANGE SERVICE COVERAGE

| Service | Descriptions |
|---|---|
| Ad server | A (computer) web server that serves, stores and maintain digital advertisement assets for websites and/or mobile services. The main functions are managing the pacing of delivery for advertising campaigns and provide reporting on campaign performance. |
| Ad ex-change | A platform of technology which could facilitate the transaction (buying and selling) of remnant mobile advertisement inventories that can be bid in real-time basis |
| Real-time analytics | Reporting and dynamic analysis that could be access by the users, when the original data are captured from inventory and/or campaign performance by the system. This approach could assist buyers and publishers to make decision with real-time insights |
| Real time bidding | Specific/designated software that provide a real-time auction of available mobile advertisement impression by accepting bids from multiple (demand) sources within a set time interval (average 100 milli-second) and delivering the advertisement to the winning bidder |
| Open RTB | Real time bidding interface which has purpose to simplify and set the requirements bar between publisher inventories suppliers (branded marketplaces, SSP, etc) |
| Programmati c | Process of simplifying the buy and sell transaction by digitally connecting buyer and sellers to bring operational and pricing efficiency through allowing the programmatic purchase of ad placements via trading platform |
| Data-mgmt platform | Platform that making the data actionable and flexible to perform a range of service, with the ability to collect, manage, optimize, organize, segment and share large amount of data |
| Software dev- kit | This kit allows mobile applications, etc to communicate with advertising software platform for the context of mobile advertisement |
| Apps interface | Determine how components of software interact with each other. Apps interface usually included in software development kit |
| Apps monetize | Monetizing income from mobile applications through app download promotion, advertising and/or other methods |
| Premium mobile | Mobile advertisement inventory that possess location data, device ID, media capabilities and mobile application data. The inventory is enriched at the impression level |
| Non-premium mobile | The pages that are not prominently featured in the context of favorite site or services. Also known as "long tail" of publishing. Despite less prominent characteristics, these pages still have a value for advertisers |
| Supply-side platform (SSP) | Also known as "sell-side platform", a technology-based platform which offer a publisher with packaging tool, pricing, brand control and reporting capabilities, to help the publisher in managing their inventory assets in a programmatic advertising world |
| Agency trading desk (ATD) | Service-based organization that centralized, which serving as a managed service layer, usually positioned between a demand side platform or other audience buying technologies and an advertising agency. Agency trading desks utilize data to assist advertisers to purchase audience-based advertisement effectively at scale across digital media |
| Whitelist | • For advertisers: site/website list that is permitted to display/place their advertisement by advertiser<br>• For publishers: list of advertisers that favorable to publisher and will permit to have access both see and bid on their inventory |

| | |
|---|---|
| First-party data | Data (user information) that is collected and stored after users visits a digital property. The process of collection and storing done by social media sites, websites, retailers, etc. |
| Third-party data | Data (user information) that is collected, repackaged and processed by companies, in order to get deeper insights through tracking and analyzing consumer characteristics or preferences. It could be used for future targeting and digital marketing purpose |
| Private ex-change | Technology-based platform that enables publisher confirm that top brand advertisers are filling publishers' premium inventory. The platform creates a controllable and efficient scheme for both advertisers and publishers to do mobile advertising |
| Inventory | a publisher's available number of advertisement/amounts of advertisement space. That could be sell to advertiser(s) |
| Audience | • For publishers: consumer characteristics of the content and/or service that provided<br>• For advertisers: advertising campaign target |
| Unstructured supplementa ry service data | GSM-based protocol that used for communicating with service provider's computers. It could be utilized for many types of services. USSD message could be reach maximum 182 alphanumeric character, through dialing a code on phone keypad as if making a phone call |

By analyzing literature review as well as expert in-depth interview, we propose the conceptual framework which displayed in Fig. 2.

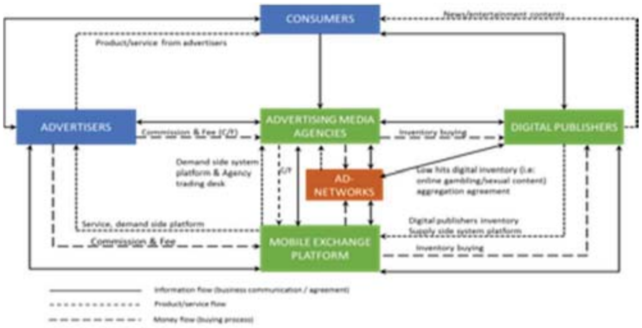


Fig. 2. Conceptual framework on digital marketing (paid advertisement) supply chain network as service e-commerce

## IV. Findings And Discussion

We propose our conceptual model based on Academic journal articles, industry reports and expert. Indepth Interview with experts which consists of two persons with with Point of discussions consists of : 1) Advertising Media agency industry landscape, 2) Digital marketing landscape in Indonesia, 3) Mobile exchange business model & services and 4) Ad-exchange business model and what aspects that make it different with Mobile Exchange. The two experts are.

First, Indonesia Mobile Exchange with its technological advancement, advance consumer analytics and supporting infrastructure that around 5 years ahead of Indonesia digital infrastructure. The infrastructure with "the Hadoop qualification" server enables the Mobile exchange not only conducting a system that operates like a stock-exchange but also supporting consumer big data analytics operations. At the beginning of its initiation, Ms C.A. explains that there are two options build the system from scratch or create a partnership with leading mobile exchange worldwide. The latest option was chosen to create "being first" in the market advantages. In the beginning of its commercial operations, Mobile exchange faces a lot of challenges, especially from advertising media agencies. Mobile exchange business model that promotes efficient and transparent pricing could harm their potential stream of revenue margin from digital marketing campaign since advertiser will pay less and digital publisher will receive more. It turns, mobile exchange strategy on training and education program for corporation especially on digital marketing and consumer big data analytics, create pressure for advertising agencies. When the clients understand about price transparency they will push their advertising media agency to use mobile exchange platform. The compromise approach to overcome advertising agency concerns is through "agency trading desk" a connection hub between advertising media agency with mobile exchange system. This connection hub has similar function with securities companies practicing transactions in stock exchange. Therefore, advertising media agencies still could have decent margin but with a more transparent context.

Second, advertising media agency has a long practice of being aggregator-middlemen which represents the interests of advertisers in buying media inventories slot, therefore media has lower bargaining power compare to the advertisers. Digital media understand the classic game and to have better business sustainability, digital media create networks and strategic alliances to have better bargaining power. Some of the alliance goes to ad-network, but more than 95.000 digital publishers are becoming the network of mobile exchange. With mobile exchange real-time bidding approach, digital publishers' bargaining power becoming stronger and to some extent protect their margin from taken by advertising media agency with un-transparent or asymmetric information digital marketing transactions.

Third, since the investment in creating a new mobile-exchange is quite high and not to mention time constraint, in short, and medium-term, it will be very rational thinking if advertising media agency has their agency trading desks. However, only a limited number of advertising media agency has its agency trading desks. If we see the future trend of e-commerce and digital advertisement based on *Techinasia* report, anticipation is needed for advertising media agencies, or the digital business will be taken slowly by mobile-exchange.

## V. Conclusion

There are two limitations within this study. First, it still in the exploratory approach and still need more expert interview from different angles especially advertisers, social media, regulator and key opinion leader. Second, only focus on paid advertising and interaction among mobile exchange-digital publishers and advertising media agencies, excluding social media that interact with paid advertising like Facebook.

Therefore, in order to get more whole-rounded point of view in future; first, more experts informants from different type of actors are needed, especially advertisers from FMCG company, management of digital publishers, management of social media platform, digital key opinion leaders, etc.; second, solid empirical future research will create better understanding toward the industry landscape and supply chain in different context.

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